Matrix functions and stability

Decidability of zero hitting by rational matrix powers

Difficulty: extreme **Importance:** broadly interesting

Status: Partially resolved

Rating rationale: Unconditional decidability in unrestricted dimension is an extreme longstanding barrier; reachability, recurrences and exact linear dynamics give broad significance.

Problem statement

Does there exist a single algorithm which, given a positive integer d, a matrix $A \in \mathbb{Q}^{d\times d}$, and vectors $u, v \in \mathbb{Q}^d$, always halts and correctly decides whether

$$u^T A^k v = 0 \qquad \text{for some integer } k \geq 0?$$

All rational inputs are represented exactly, for example by binary integers for numerators and nonzero denominators. There is no a priori bound on d, on the entries, or on the hitting time k. The output sought is a yes/no answer; numerical near-zero detection does not suffice.

This is the Skolem problem in its matrix-power form. Equivalently, one is given rational coefficients and initial values for a finite-order scalar linear recurrence and asks whether one of its terms vanishes. These equivalent formulations form one catalog entry.

The matrix formulation asks whether a discrete linear system reaches a specified hyperplane. It is directly relevant to exact observability, reachability, and verification of linear dynamics, and exposes a fundamental limit of computations with powers of matrices.

References

1. P. Bacik, T. Karimov, F. Luca, J. Nieuwveld, J. Ouaknine, D. Purser and J. Worrell, *A survey of the Skolem and Positivity Problems for linear recurrence sequences*, submitted (2026), §1, Problem 2, and the discussion of linear dynamical systems; §5. Author-hosted PDF.
2. F. Luca, J. Ouaknine and J. Worrell, *Conjectural Decidability of the Skolem Problem*, arXiv:2607.15510v1 (2026), §1. Full preprint.

Status check (2026-09-10): the 2026 survey still poses the general problem. The July 16 preprint proves conditional decidability under a strengthened prime-gap conjecture and an unconditional density-one universal Skolem-set result; it does not supply an unconditional algorithm on all inputs. Its current arXiv record is v1. Searches for “Skolem Problem decidability solution 2026”, including September developments, found special-family and positive-characteristic results but no general rational-input resolution. This is a dated, bounded status check.

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Independently rechecked Luca–Ouaknine–Worrell, §1, its current v1 record, and the 2026 survey’s author abstract. The order-at-most-four decidability result gives a substantive exact subclass (in particular, $d \leq 4$ here), justifying Partially resolved; conditional decidability and density-one sets do not settle unrestricted inputs. Later Skolem-decidability searches found no unconditional general resolution. Both ratings are retained.